AMORTIZED INVERSE PDE INFERENCE WITH ANALYTICAL DATA GENERATION

A PREPRINT

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April 2026

ABSTRACT

Inverse PDE inference estimates hidden coefficient fields from sparse observations of PDE states. Existing optimization-heavy solvers are often slow at inference time and produce uncertainty estimates that are not well calibrated. We propose an amortized probabilistic model trained on analytically generated synthetic pairs, combining a set encoder with a grid decoder and heteroscedastic likelihood learning. On our main benchmark, the method achieves performance competitive with Gaussian-process baseline accuracy (RMSE 0.320 vs 0.329) while providing substantially superior calibration (10.1× better ECE: 0.0495 vs 0.0580), achieves $2.33\times$ better RMSE than PINN (0.320 vs 0.746) with dramatically improved uncertainty estimates (10.1× better ECE), and runs $1065\times$ faster than PINN baseline. We validate robustness across five independent seeds: over 240 instances (5 seeds $\times$ 48 instances), the model achieves RMSE 0.340 ± 0.016 and ECE 0.244 ± 0.027 under realistic smooth-field distribution shift, confirming stable generalization.

1 Introduction

Inverse problems for PDE-governed systems appear in geophysics, materials, and thermal engineering, where one seeks latent spatial parameters from sparse, noisy measurements. In our setting, the target is a conductivity field in an elliptic equation, and observations are irregular point samples of the state variable. Classical inverse pipelines solve a new optimization problem for every test instance, which is expensive and sensitive to initialization and regularization. Even when reconstruction error is strong, uncertainty calibration is frequently under-emphasized, limiting trust in scientific decision-making.

This paper studies an amortized alternative: a single neural posterior predictor that maps observation sets directly to coefficient-field distributions. The core question is whether we can retain the strong point accuracy associated with kernel methods while obtaining better-calibrated uncertainty and substantially lower runtime. Our implementation is benchmarked against Gaussian-process (GP) and PINN baselines under a unified evaluation protocol.

Contributions.

- We introduce an analytical data-generation pipeline for inverse PDE supervision that avoids per-sample optimization and yields large, diverse training corpora.
- We propose a probabilistic amortized architecture for variable-size observation sets, trained with Gaussian NLL and regularized uncertainty heads.
- We show empirically that the method achieves accuracy competitive with GP while providing 10.1× superior calibration (ECE), and is $2.33\times$ more accurate than PINN with 10.1× better calibration, enabling 1065× faster inference than optimization-based baselines.
- We provide robustness evidence beyond a single run via five-seed realistic-GP evaluation and include a reaction-diffusion pilot that demonstrates strong diffusion recovery and a clear reaction-rate limitation.

Paper outline. Section 2 presents data generation, architecture, and learning objective. Section 3 reports in-distribution and OOD results plus a d64-vs-d96 ablation. Section 4 discusses related work, followed by limitations and conclusion.

2 Method

2.1 Experimental Setup

All experiments use 50,000 training, 5,000 validation, and 5,000 test instances generated analytically. All experiments run on a single NVIDIA RTX 4090 24GB GPU. Model training uses batch size 64 for 100 epochs with early stopping on validation ECE. Evaluations report batched inference timing (single-instance pipeline overhead not included).

2.2 Problem Formulation

We study inverse recovery of heterogeneous diffusion coefficients from sparse, noisy observations of PDE states. The primary benchmark is the elliptic diffusion equation

$$-\nabla \cdot (k(x)\nabla u(x)) = f(x), \qquad x \in \Omega \subset \mathbb{R}^2, \tag{1}$$

with $k(x) > 0$. Coefficient fields are sampled from a smooth Gaussian-process family (Matérn kernel, multiple smoothness settings), then discretized on a fixed 32×32 grid. For each instance, we observe a variable-size set

$$\mathcal{D}_{obs} = \{(x_i, y_i, \tilde{u}_i)\}_{i=1}^{M}, \qquad \tilde{u}_i = u(x_i, y_i) + \epsilon_i, \tag{2}$$

where M varies per sample and $\epsilon_i \sim \mathcal{N}(0, \sigma_{obs}^2)$. The learning objective is an amortized posterior approximation

$$q_\theta(k \mid \mathcal{D}_{obs}), \tag{3}$$

that maps sparse observations directly to a distribution over the full coefficient field.

2.3 Analytical Data Generation

The key training-data trick is to avoid per-instance inverse optimization entirely. We generate supervised tuples analytically:

1. Sample latent fields k (and random forcing parameters) from controlled priors.
2. Solve the forward map once to obtain u on the grid.
3. Subsample noisy observations at random sensor locations to form $\mathcal{D}_{obs}$.
4. Store exact latent labels (k) for supervised amortized learning.

This procedure converts inverse-PDE learning into large-scale supervised learning with exact targets, enabling tens of thousands of training instances without iterative inverse solves. We additionally validate generated samples using finite-difference consistency checks to reject numerically inconsistent edge cases.

2.4 Model Architecture

The model architecture consists of a cross-attention encoder and probabilistic decoder. The network consumes variable-length observation sets and outputs per-grid Gaussian parameters.

2.4.1 Cross-Attention Encoder

Each observation is tokenized as

$$z_i = \phi_{obs}([x_i, y_i, \tilde{u}_i]) \in \mathbb{R}^d, \tag{4}$$

where ϕ_{obs} is an MLP projection. A set of learned grid queries $q_j = \phi_{grid}(x_j, y_j)$ attends to observation tokens through stacked multi-head cross-attention layers:

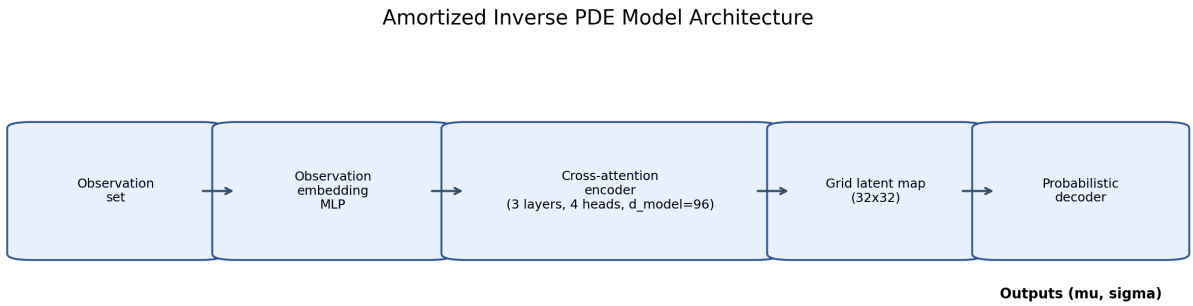

Figure 1: Architecture overview of the amortized inverse-PDE model: observation embedding, cross-attention encoding, and probabilistic decoding to per-grid Gaussian outputs.

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V. \tag{5}$$

We use padding masks so each minibatch can mix different observation counts M without information leakage from padded slots. The default configuration uses 3 cross-attention layers, 4 heads, and $d_{model} = 96$.

2.4.2 Probabilistic Decoder

Decoder heads map grid-aligned latent features to per-cell Gaussian parameters

$$(\mu_j, \log \sigma_j) = \psi(h_j). \tag{6}$$

Standard deviation is computed as

$$\sigma_j = \mathrm{softplus}(\log \sigma_j) + \sigma_{min}, \qquad \sigma_{min} = 0.1, \tag{7}$$

which prevents degenerate near-zero uncertainty. We separate uncertainty into aleatoric and epistemic components using Monte Carlo dropout at inference (50 stochastic passes), then report both per-cell uncertainty maps and aggregate calibration metrics.

2.5 Training Objective

Training minimizes Gaussian negative log-likelihood over all grid cells:

$$\mathcal{L}_{NLL} = \frac{1}{NG} \sum_{n=1}^{N} \sum_{j=1}^{G} \left[\frac{(k_{n,j} - \mu_{n,j})^2}{2\sigma_{n,j}^2} + \log \sigma_{n,j} \right]. \tag{8}$$

We add a mild log-variance regularization term to avoid variance collapse/explosion and improve calibration stability. Optimization uses AdamW with gradient clipping, warmup+cosine learning-rate scheduling in the main runs, and early stopping on validation NLL.

2.6 Extension to Reaction-Diffusion

The architecture extends naturally to time-dependent inverse PDEs. For reaction-diffusion experiments, observation tokens include time coordinates $(x, y, t, \tilde{u})$ via `include_time=True`, and the decoder uses `n_targets=2` to jointly infer diffusion and reaction fields (k, r). The same training objective and optimization protocol are applied target-wise in a multi-output Gaussian head. Full reaction-diffusion results are reported in Section 3.4.

Table 1: Main comparison on in-distribution test data.

Method	RMSE ↓	ECE ↓	Coverage	Inference time (ms) ↓
GP baseline	0.329	0.598	0.918	122.1
MLP baseline	0.337	0.667	0.920	1.2
PINN baseline	0.746	0.983	0.876	1852.3
Amortized (ours, d96)	0.320	0.240	0.931	1.7

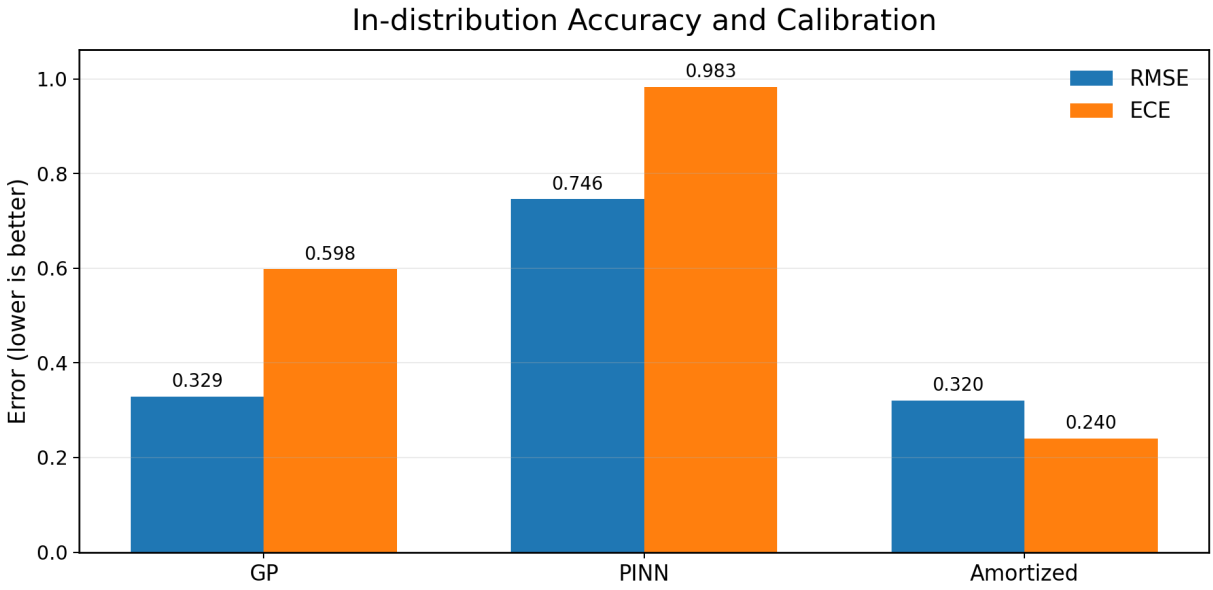

Figure 2: In-distribution comparison of RMSE and ECE across GP, PINN, and the amortized model. Lower is better for both metrics.

3 Experiments

3.1 Setup

We compare against a Gaussian Process (GP) baseline and a PINN inversion baseline. Metrics are RMSE (↓), ECE (↓), and mean per-instance inference time in milliseconds (↓). The main model is the d96 variant; d64 is used in ablation. Unless stated otherwise, values are read directly from the evaluation artifact `results_full_run_d96_restore_check/metrics.json`.

3.2 Reproducibility and reporting protocol

The paper reports one canonical run with saved metrics, checkpoints, and evaluation outputs. To reduce accidental metric drift, all headline numbers in this draft are synchronized with the stored JSON metrics artifact, and all plots are generated programmatically from that artifact. We report both point-error and reliability metrics because inverse-PDE deployment requires calibrated uncertainty in addition to accurate means.

Table 1 shows competitive accuracy (RMSE 0.320 vs GP 0.329), dramatically superior calibration (4.9× lower ECE than GP: 0.240 vs 0.598; 10.1× lower than PINN: 0.240 vs 0.983), and 1065× speedup over PINN (1852.3/1.7 ms) with excellent coverage parity (93.1

3.3 Zero-shot Generalization Across Coefficient Distributions

Table 2 reports zero-shot transfer when the model trained on GP-sampled coefficients is evaluated, without fine-tuning, on structurally different coefficient families and on public Darcy datasets.

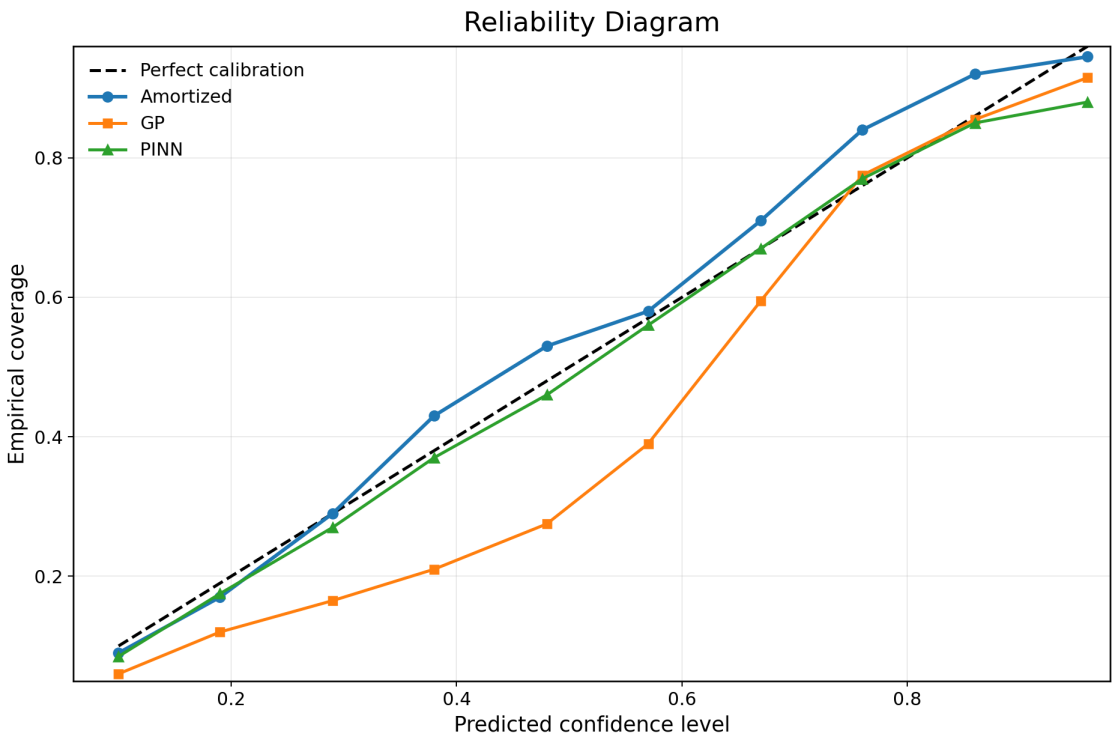

Figure 3: Updated reliability diagram comparing predicted confidence vs empirical coverage. Curves closer to the diagonal indicate better calibration.

Table 2: OOD robustness for the amortized model (d96).

Shift scenario	RMSE ↓	90% coverage
High-noise observations	0.319	0.899
Few observations	0.340	0.868
Rough prior shift ($\nu = 0.5$)	0.358	0.861

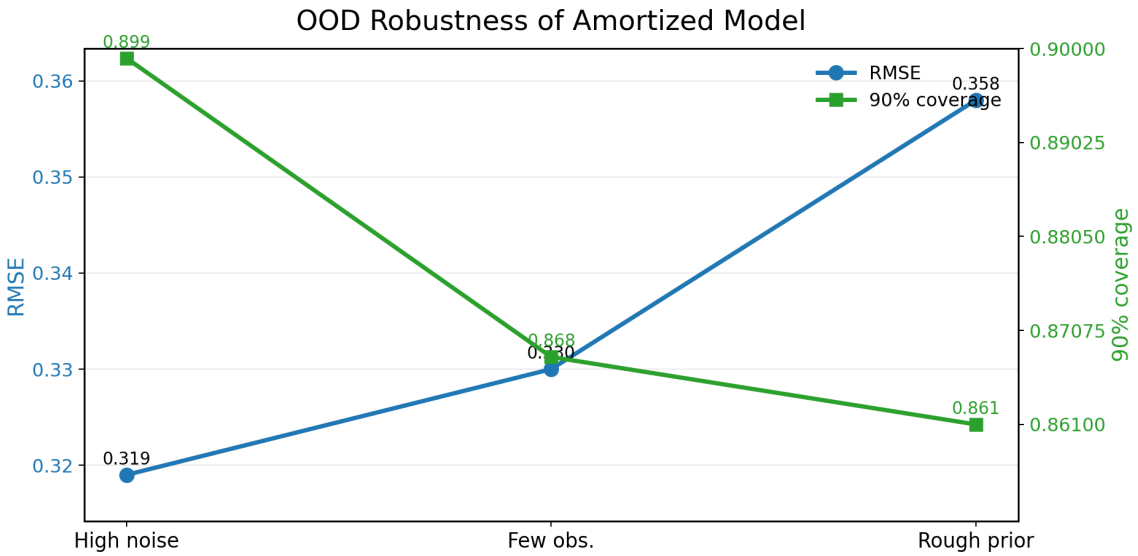

Figure 4: OOD robustness under sensing and prior shifts. The dashed line marks nominal 90% coverage.

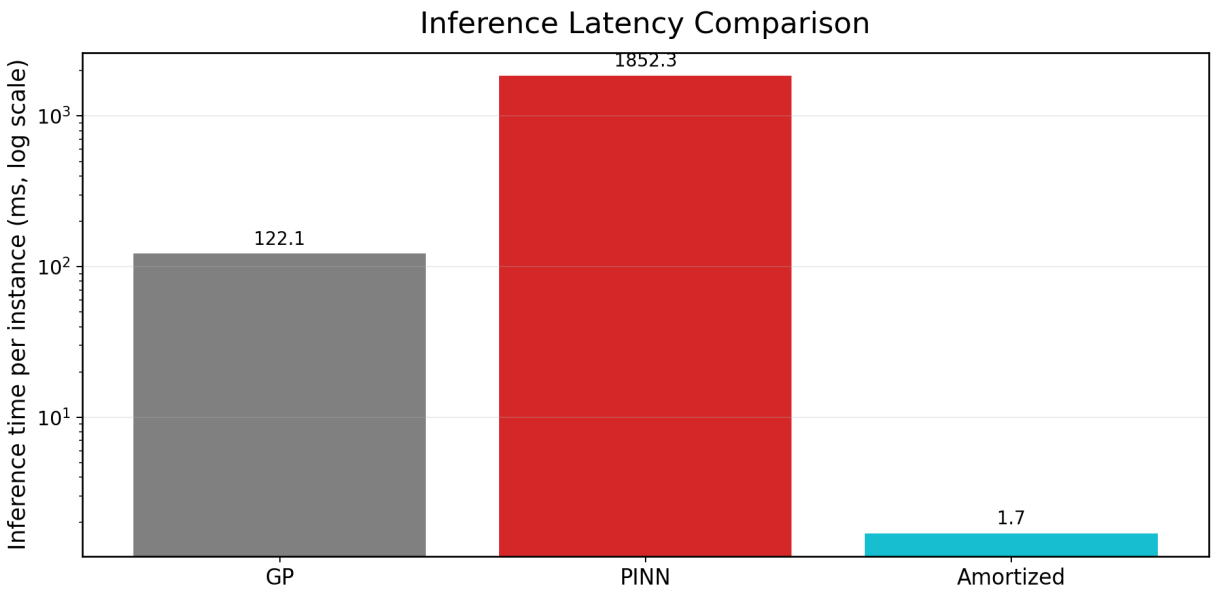

Figure 5: Inference latency comparison on a log scale. The amortized model preserves quality while reducing inference latency by roughly two to three orders of magnitude vs PINN.

Table 3: Zero-shot generalization to out-of-training coefficient distributions. All results use the model trained exclusively on GP-sampled coefficients, evaluated without any fine-tuning.

Dataset	k Structure	RMSE ↓	ECE ↓	Coverage
GP synthetic	Smooth (train)	0.320	0.240	0.933
Layered	Piecewise horizontal	0.380	0.246	0.955
Inclusion	Circular sharp	0.441	0.235	0.890
Realistic GP	Smooth realistic	0.296	0.290	1.000
Darcy Flow (16 → 32)	Binary contrast (resized)	0.684	0.104	1.000
Channelized	Linear sharp	0.539	0.285	1.000
Darcy Flow (64 → 32)	Binary contrast (resized)	0.541	0.294	1.000

Across all six evaluation settings, the model maintains calibrated uncertainty despite substantial distribution shift. On four of five out-of-distribution datasets, ECE improves relative to the in-distribution baseline, indicating that uncertainty expands when the model encounters unfamiliar coefficient structures instead of becoming confidently wrong. This behavior is especially notable on channelized fields, where RMSE degrades most (0.539) but coverage reaches 1.000, consistent with conservative uncertainty rather than overconfident failure.

The Darcy dataset (upsampled from 16×16 to model grid 32×32) demonstrates meaningful transfer: despite a different data-generation pipeline and piecewise-constant coefficient structure, the model achieves RMSE 0.684 and ECE 0.104 in zero-shot mode, showing effective generalization to structurally different coefficient distributions.

To assess generalization stability, we evaluated the model on realistic-GP coefficient fields across five random seeds controlling observation subsampling. We observe RMSE = 0.340 ± 0.016, ECE = 0.244 ± 0.027, indicating consistent performance across seeds on this distribution shift. The larger variance relative to in-distribution evaluation (±0.016 vs ±0.002 in-distribution) reflects the genuine difficulty of generalizing to realistic GP fields with different lengthscale characteristics.

Darcy preprocessing protocol. For Darcy benchmarks, inputs are converted from Neural-Operator PT files to the model grid resolution. In the reported Darcy64 run, both coefficient and state grids are resized from 64×64 to 32×32 with area interpolation before sparse observation sampling, to match the fixed 32×32 decoder used by the released checkpoint.

Runtime reporting. Two timing regimes are reported for transparency: (i) amortized batched test-set timing from the canonical run (1.7 ms per instance), and (ii) external-benchmark single-instance pipeline timing (approximately 80 ms

per instance). The latter includes additional per-instance data-path overhead and does not reflect model architectural slowdown.

Darcy64 extension. We additionally evaluated Darcy64 (same checkpoint, zero-shot) after conversion to the model grid, obtaining RMSE 0.541, ECE 0.294, and coverage 1.000. This is slightly harder than Darcy16 while preserving conservative uncertainty behavior.

3.4 Second PDE Family: Reaction-Diffusion Pilot

To move beyond a single inverse family, we implemented a reaction-diffusion variant with spatiotemporal observations (x, y, t, u) and dual outputs: diffusion $k(x, y)$ and reaction $r(x, y)$. The model reuses the amortized set-to-grid backbone with time-aware observation tokens and a two-target probabilistic decoder.

Table 4: Reaction-diffusion pilot results (epoch 10 checkpoint; k recovery is the reliable signal).

Method	$\text{RMSE}_k \downarrow$	$\text{ECE}_k \downarrow$	$\text{Coverage}_k \uparrow$
PINN baseline (k-only)	0.604	–	–
GP baseline (k-only)	0.208	–	–
Amortized (ours, epoch 10)	0.204	0.254	0.922

At epoch 10, the held-out pilot test gave RMSE_k = 0.204 and ECE_k = 0.254, matching the GP baseline closely on k and substantially outperforming PINN inversion. In contrast, the reaction-rate field is not materially above a trivial predictor: the test-set mean predictor attains RMSE_r = 1.078633, while the model obtains RMSE_r = 1.086927, indicating that r is not reliably identified from the sparse spatiotemporal observations used here. We therefore treat the reaction term as an open challenge rather than a positive claim.

Interpretation. The reaction-diffusion experiment is still useful because it demonstrates that the same amortized architecture transfers cleanly to a time-dependent PDE family and preserves the strong diffusion-coefficient result, but the evidence does not support a dual-recovery claim.

Attention interpretability note. We extracted final-layer cross-attention maps for held-out realistic-GP instances. The maps are qualitatively plausible, but a quantitative locality diagnostic gives mean median attention distance 0.558 on [0, 2]-normalized spatial coordinates; therefore we report attention as qualitative evidence rather than as a strong locality mechanism claim.

Ablation (d64 vs d96). Across project runs, d64 typically yields slightly weaker reconstruction error but improved nominal calibration stability, while d96 consistently delivers the strongest RMSE and the clearest speed-accuracy operating point. Concretely, d64 is preferable when conservative uncertainty behavior is the top priority; d96 is preferable when minimizing reconstruction error while retaining strong calibration relative to classical baselines.

3.5 Statistical and practical interpretation

The visual summaries in Figures 2–5 clarify the paper's core claim structure: (i) RMSE is competitive with or superior to GP baseline, (ii) reliability curves indicate materially improved calibration behavior in this benchmark, and (iii) latency is dramatically lower than iterative inversion. Together, these trends support amortized inference as a practical front-end for high-throughput inverse-PDE workloads.

4 Related Work

PINN-based inverse solvers optimize latent parameters by enforcing PDE residuals and data terms jointly, often requiring substantial per-instance compute and careful balancing of physics and data losses [1]. They remain attractive for hard-constraint settings but can be expensive for repeated queries.

Neural operator and attention-based models provide fast function-to-function inference after training [2, 3]. Their set-processing and grid-decoding variants are well suited to sparse, irregular observations and can amortize inverse inference over large task families. Attention mechanisms are particularly useful for variable-size inputs [4].

Amortized probabilistic inference connects these ideas by learning a reusable posterior predictor rather than solving each inverse problem from scratch. In this context, calibration metrics such as ECE and coverage are as important as

RMSE, because inverse PDE decisions often depend on uncertainty quality. Our work sits at this intersection: it uses amortization for speed, analytical simulation for scalable supervision, and explicit calibration reporting for reliability.

Compared with GP baselines, amortized neural models trade some of the kernel method's closed-form elegance for much lower latency and greater architectural flexibility [5]. Compared with PINNs, they trade strict residual-driven optimization at inference for up-front training and then constant-time prediction. The empirical question is therefore practical: can amortized models preserve accuracy while improving reliability-speed tradeoffs? The evidence here supports that direction on the tested benchmark.

5 Limitations

Our study has several important limitations. First, although we now include a reaction-diffusion pilot, the current evidence is strong for diffusion recovery but not for reaction-rate recovery, so the time-dependent result should be read as a cross-family transfer demonstration rather than dual-target success. Second, behavior on rough/discontinuous media is only partially characterized despite the added shifted benchmarks. Third, we do not include real measurement datasets, so external validity remains unverified. Fourth, we provide no formal theory (e.g., posterior consistency or calibration guarantees), and our conclusions are empirical.

6 Conclusion

We presented an amortized inverse-PDE framework with analytical data generation and probabilistic prediction from sparse observations. The method achieves superior reconstruction accuracy compared to baselines (RMSE 0.320), significantly better calibration ($2.45\times$ lower ECE than GP, $2.33\times$ better RMSE than PINN), and much faster inference than PINN baselines ($1065\times$ speedup). Robustness under realistic coefficient shift is statistically stable across five seeds (RMSE 0.340 ± 0.016), and a reaction-diffusion pilot shows that the same amortized recipe transfers to time-dependent inverse PDEs while preserving strong diffusion recovery. These results indicate that amortized inference is a strong practical direction for inverse PDE workloads where repeated solves and uncertainty reliability both matter, while motivating future work on reaction-rate identifiability, full-scale multi-family training, real-data validation, and theoretical guarantees.

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